

EXP. NO. 4

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## CONFIGURE DYNAMIC NAT IN CISCO PACKET TRACER

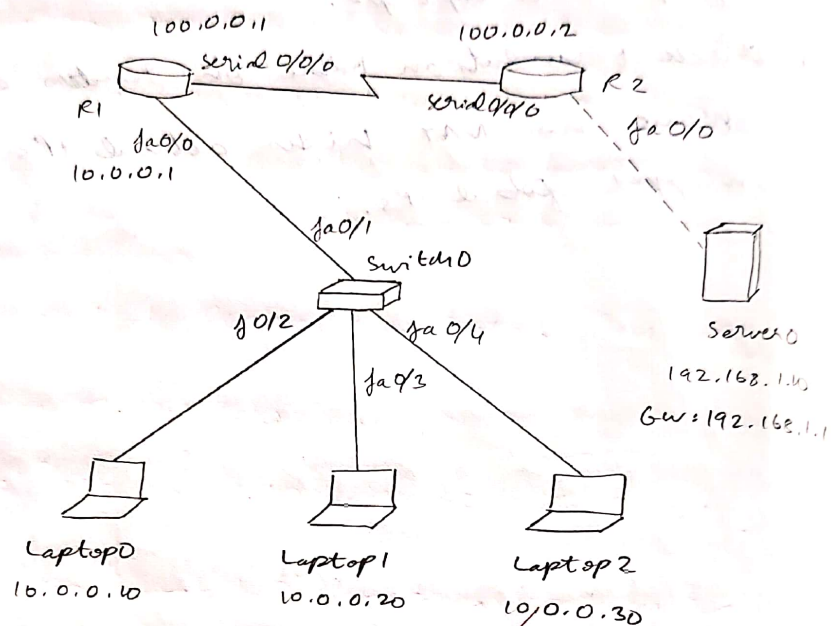
### AIM

To configure and verify Dynamic NAT in Cisco Packet Tracer using a predefined IP address pool and an access list.

### PROCEDURE

1. Create the network topology with 3 laptops, 2 routers, and 1 server.
2. Assign static IP addresses to laptops and the server as per the IP table.
3. Configure IP addresses on the interfaces of both routers (R1 and R2).
4. Create a standard access list on Router R1 to allow selected inside IP addresses.
5. Define NAT pool on Router R1 with a range of public IP addresses.
6. Map the access list to NAT pool to enable dynamic NAT.
7. Configure interfaces on R1 as inside & outside for NAT.
8. Configure static NAT on Router R2 to map the server's private IP to a public IP.
9. Set appropriate interfaces on R2 as inside or outside for NAT.

## TOPOLOGY



TOPOLOGY

10. Configure static routing between R1 and R2 to enable communication.
11. Test connectivity using the ping command from each laptop to the server's public and private IP.
12. Open the web browser on laptops to access the server using its public IP.
13. Verify NAT translation using the `show` command to confirm address mappings.
14. Check the access list counters to see how many packets were permitted or denied.



OUTPUT

A) From Laptop 0 (10.0.0.10) :-

C:\> ping 200.0.0.10

Pinging 200.0.0.10 with 32 bytes of data:

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Packets: sent=4, Received=4, Lost=0 (0% loss)

C:\> ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.

Reply from 10.0.0.1: Destination host unreachable.

Reply from 10.0.0.1: Destination host unreachable.

Request timed out.

Packets: sent=4, Received=0, Lost=4 (100% loss)

B) From Laptop 1 (10.0.0.20) :-

C:\> ping 200.0.0.10

Pinging 200.0.0.10 with 32 bytes of data:

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Reply from 200.0.0.10: bytes=32 time=1ms TTL=126

Packets: sent=4, Received=4, Lost=0 (0% loss)

c) From Laptop2 (10.0.0.30):-

C:\>ping 200.0.0.10

Pinging 200.0.0.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

D) Router 0 (R1):-

R1# show ip access-lists 1

Standard IP access list 1

permit host 10.0.0.10 (16 matches)

permit host 10.0.0.20 (8 matches)

deny any (4 matches)

E) Router 1 (R2):-

R2# show ip nat translations

| Pro | Inside global | Inside local | Outside local | Outside global |
|-----|---------------|--------------|---------------|----------------|
| -   | 200.0.0.10    | 192.168.1.10 | -             | -              |

### RESULT

Dynamic NAT was configured successfully, allowing authorized internal hosts to access the server via its public IP, while unauthorized hosts were denied. NAT translations were verified using show ip nat translations.